

Salvador Balkus

Curriculum Vitae

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Research Interests

Statistical Causal machine learning, semiparametrics, spatial and network data
Scientific Environmental health, pollution, public policy

Education

2022—2027 **Doctor of Philosophy in Biostatistics**, *Harvard University*
Advised by Professor Nima Hejazi. Thesis: Nonparametric Causal Inference for Environmental Health
2018—2022 **Bachelor of Science in Data Science**, *University of Massachusetts Dartmouth*
Minors: Mathematics, Economics. GPA: 4.0/4.0; Summa Cum Laude

Experience

2026—Present **Graduate Student Researcher**, *Implementation Science Center for Cancer Control Equity*,
Harvard University, Cambridge, MA
Collaborated with clinicians and domain experts to perform statistical analysis of colorectal cancer
screening data from community health centers using targeted machine learning.
2022—Present **Graduate Student Researcher**, *NSH Lab*,
Harvard University, Cambridge, MA
Designed and implemented computational improvements to statistical learning methodology for causal
inference, including open-source software.
2022—Present **Graduate Student Researcher**, *National Studies on Air Pollution and Health*,
Harvard University, Cambridge, MA
Developed new causal inference methods for spatial and network data with continuous treatments
(initially through National Institutes of Health Environmental Statistics Training Grant Program).
2020—2022 **Research Assistant**, *Computational Statistics & Data Science Lab*,
University of Massachusetts Dartmouth, North Dartmouth, MA
Implemented algorithms for smart health data analysis and dietary data harmonization.

Research Fellowships

2022 **Graduate Research Fellowship**, *National Science Foundation*

Awards

2025 **Early Career Travel Award**, *Joint Statistical Meetings*, Nashville, TN
2024—2025 **Certificate of Distinction in Teaching**, *BST 232: Methods I*, Harvard University
2022 **Best Analysis and Visualization**, *SEMASS DataFest*, American Statistical Association
2021 **Travel Award**, *Joint Mathematics Meetings*, American Mathematical Society

Publications and Working Papers

Statistical Methodology

- [1] **S. V. Balkus**, C. Testa, and N. S. Hejazi, “A Riesz representer perspective on Targeted Learning”, *Accepted to special issue of the International Journal of Biostatistics*, 2026.
- [2] **S. V. Balkus**, S. W. Delaney, and N. S. Hejazi, “The causal effects of modified treatment policies under network interference”, *Journal of the Royal Statistical Society Series B: Statistical Methodology*, Mar. 2026, ISSN: 1467-9868. DOI: 10.1093/jrsssb/qkag052.
- [3] **S. V. Balkus** and N. S. Hejazi, “CausalTables.jl: Simulating and storing data for statistical causal inference in Julia”, *Journal of Open Source Software*, vol. 10, no. 106, p. 7580, 2025. DOI: 10.21105/joss.07580.

Preprints

- [4] E. Tong and **S. V. Balkus**, “Linear models for causal inference under network interference”, *arXiv preprint arXiv:2603.29168*, 2026.
- [5] **S. V. Balkus**, H. Laith, and N. S. Hejazi, “On the use of cross-fitting in causal machine learning with correlated units”, *arXiv preprint arXiv:2601.10899*, 2026.

Works in Preparation

- [6] **S. V. Balkus** and N. S. Hejazi, “Quickly approximating a learning algorithm while preserving its statistical rate”, *In preparation*, 2026.
- [7] **S. V. Balkus** and N. S. Hejazi, “Function estimation under network interference”, *In preparation*, 2026.
- [8] C. Jiang, **S. V. Balkus**, and N. S. Hejazi, “A unified projection theory for conditional and coarsened mean functionals”, *In preparation*, 2026.

Applied Computer Science

- [9] **S. V. Balkus** and D. Yan, “Improving short text classification with augmented data using GPT-3”, *Natural Language Engineering*, pp. 1–30, 2023. DOI: 10.1017/s1351324923000438.
- [10] **S. V. Balkus**, H. Fang, and H. Wang, “Federated fuzzy clustering for longitudinal health data”, *2022 IEEE/ACM Conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE)*, pp. 128–132, 2022.
- [11] **S. V. Balkus**, H. Wang, B. D. Cornet, C. Mahabal, H. Ngo, and H. Fang, “A survey of collaborative machine learning using 5G vehicular communications”, *IEEE Communications Surveys & Tutorials*, vol. 24, no. 2, pp. 1280–1303, 2022. DOI: 10.1109/comst.2022.3149714.
- [12] V. S. Gurugubelli, H. Fang, J. M. Shikany, **S. V. Balkus**, J. Rumbut, H. Ngo, H. Wang, J. J. Allison, and L. M. Steffen, “A review of harmonization methods for studying dietary patterns”, *Smart Health*, vol. 23, p. 100263, 2022. DOI: 10.1016/j.smhl.2021.100263.
- [13] **S. V. Balkus**, H. Fang, J. Rumbut, A. Moormann, and E. Boyer, “A multi-level biosensor-based epidemic simulation model for COVID-19”, *IEEE Internet of Things Journal*, pp. 1–1, 2021. DOI: 10.1109/jiot.2021.3127804.
- [14] **S. V. Balkus**, J. Rumbut, H. Wang, and H. Fang, “An adaptive and dynamic biosensor epidemic model for COVID-19”, in *2020 IEEE 21st International Conference on Information Reuse and Integration for Data Science (IRI)*, IEEE, Aug. 2020. DOI: 10.1109/iri49571.2020.00051.

Presentations

Conferences

- [1] **S. V. Balkus**, H. Laith, and N. S. Hejazi, “Cross-fitting in causal machine learning with correlated units”, in *Joint Statistical Meetings*, Aug. 2026.
- [2] **S. V. Balkus**, C. Testa, and N. S. Hejazi, “A Riesz representer perspective on Targeted Learning”, in *European Causal Inference Meeting*, Apr. 2026.
- [3] **S. V. Balkus** and N. S. Hejazi, “Lasso rates hold even after randomly discarding most of the predictors”, in *Joint Statistical Meetings*, Aug. 2025.
- [4] **S. V. Balkus**, R. C. Nethery, S. W. Delaney, and N. S. Hejazi, “Nonparametric network causal inference for continuous exposures in mobile source air pollution”, in *American Causal Inference Conference*, May 2024.
- [5] **S. V. Balkus**, R. C. Nethery, S. W. Delaney, and N. S. Hejazi, “Assumption-lean causal inference for mobile source air pollution”, in *ASA Boston Chapter Student Research Symposium on Statistics and Data Science*, Apr. 2024.
- [6] **S. V. Balkus**, “Language models that teach themselves: Augmenting training data for topic classification using GPT-3”, in *ASA Boston Chapter Student Research Symposium on Statistics and Data Science*, Apr. 2022.
- [7] N. Pai, **S. V. Balkus**, and T. Zeng, “Multi-hypothesis tracking of space objects and targets”, in *AMS Joint Mathematics Meetings (JMM) Poster Session*, Apr. 2022.
- [8] **S. V. Balkus**, “Multi-level biosensor-based epidemic forecasting in small areas”, in *Joint Statistical Meetings*, American Statistical Association, Aug. 2021.

Invited Talks

- [9] **S. V. Balkus**, C. Testa, and N. S. Hejazi, “TMLE made easy with Riesz representers”, in *Harvard Biostatistics Student Seminar*, Sep. 2026.
- [10] **S. V. Balkus**, S. W. Delaney, and N. S. Hejazi, “Induced modified treatment policies for estimating effects of continuous exposures under interference”, in *University of Washington TGIF Reading Group*, May 2026.
- [11] S. Woodward and **S. V. Balkus**, “Spatial confounding and interference in environmental health studies”, *BUSPH-HSPH CAFE Research Coordinating Center Webinar*, Feb. 2026.
- [12] **S. V. Balkus**, S. W. Delaney, and N. S. Hejazi, “The causal effects of modified treatment policies under network interference”, in *Space-Time Causality Reading Group*, <https://spacetimecausality.github.io/>, Sep. 2025.
- [13] **S. V. Balkus**, “Speeding up the Highly Adaptive Lasso”, in *Harvard Biostatistics Student Seminar*, Mar. 2025.
- [14] **S. V. Balkus**, “Statistics in Julia: Is it right for you?”, in *Harvard Biostatistics Student Seminar*, Dec. 2024.
- [15] **S. V. Balkus**, “Lunchtime Computing: Basics of AWS SageMaker”, in *Center for Scientific Computing and Visualization Research*, University of Massachusetts Dartmouth, Feb. 2021.
- [16] **S. V. Balkus**, “Lunchtime Computing: Getting started with Git and GitHub”, in *Center for Scientific Computing and Visualization Research*, University of Massachusetts Dartmouth, Feb. 2020.

Mentoring

- 2026—Present **NSH Lab**, *Jolene Cao*, Project: “Simple Riesz learners for causal inference”
- 2025—2026 **NSH Lab**, *Hasan Laith*, Project: “On the use of cross-fitting for causal machine learning in dependent-data settings”
- 2025—2026 **National Studies on Air Pollution and Health**, *Eric Tong*, Project: “Linear models under network interference”
- Summer 2025 **Harvard StatStart**, *Maria Huang, Moulik Pagarani, and Ian Ruben Rivera Rodriguez*, Project: “Mapping the Legacy of Redlining”

Teaching

Fellowships

- Fall 2026 **Teaching Fellow**, *BST 220: Applied Generalized Linear Models*, Harvard University
For master’s students in biostatistics and across fields of public health to learn practical aspects of linear modeling; held office hours, graded, and taught labs.
- Fall 2024 **Teaching Fellow**, *BST 232: Methods I*, Harvard University
For PhD and master’s students to learn theory of linear modeling, hypothesis testing, computational methods, and Bayesian statistics; held office hours, graded, and taught & designed labs.
- Summer 2024 **Curriculum Fellow**, *BST 232: Methods I*, Harvard University
Aided in curriculum redesign to split Methods into two courses; created new, improved labs and instructional materials for Methods I.
- Fall 2023 **Teaching Fellow**, *BST 232: Methods*, Harvard University
For PhD and master’s students to learn theory of linear models, computational methods, categorical data analysis, and generalized linear models; held office hours, graded, and taught & redesigned labs.

Programs

- 2024—2025 **Qualifying Examination Tutor**, Harvard University
Taught refresher on regression and optimization topics and led review sessions for first-year Biostatistics PhD Qualifying Examination.
- July 2025 **Research Project Instructor**, *Harvard StatStart*, Harvard University
For underserved high-school students in the Boston area to be introduced to statistics; designed month-long high-school-level data analysis project in R and mentored research team of three.
- June 2024 **Workshop Instructor**, *Replicathon, IQ BIO REU*, University of Puerto Rico Río Piedras
Co-led 2-day, in-person workshop in Puerto Rico for undergraduate students on data analysis in R.

Service and Extracurricular

Ad-hoc Reviewer

Journal of the Royal Statistical Society Series B (Statistical Methodology), Journal of Causal Inference [3x], The R Journal, Journal of Open Source Software [4x], Natural Language Processing, BMC Medical Informatics and Decision Making

- 2024 —Present **Co-Chair**, *Biostatistics Student Committee*, Harvard University
Led peer mentoring program, qualifying exam tutoring program, social event planning, and resource organization for 70+ PhD students.
- 2022 —2026 **Graphic Designer and Blog Contributor**, *Science in the News*, Harvard University
Designed infographics and wrote blog posts explaining scientific topics to a lay audience.
- 2025 **Session Chair**, *Causal Discovery*, *Joint Statistical Meetings*, Nashville, TN
- 2023 —2024 **Committee Chair**, *Biostatistics Peer Mentoring Program*, Harvard University
Supported class of 12 incoming students by serving as peer mentor, creating instructional material, acquiring event funding, and planning seminar series.

Scientific Blog Contributions

- [1] H. Farnsworth (Author) and **S. V. Balkus (Illustrator)**, “Take a deep breath: How stress reduction may influence cancer risk”, *Harvard Science in the News*, Jul. 2026.
- [2] E. Crowley-Dolen (Author) and **S. V. Balkus (Illustrator)**, “Suzetrigine and the shift toward non-opioid pain management”, *Harvard Science in the News*, Oct. 2025.
- [3] A. Bogatch (Author) and **S. V. Balkus (Illustrator)**, “Pregnancy causes astonishing brain changes why haven’t we noticed this before?”, *Harvard Science in the News*, May 2025.
- [4] S. Renauld (Author) and **S. V. Balkus (Illustrator)**, “Gene editing to treat disease: A success story in sickle cell disease”, *Harvard Science in the News*, May 2024.
- [5] H. Blume (Author) and **S. V. Balkus (Illustrator)**, “Liquid-within-liquid compartments: A ground-breaking discovery in biology”, *Harvard Science in the News*, Jun. 2023.
- [6] **S. V. Balkus (Author and Illustrator)**, “How do scientists know whether to trust their results?”, *Harvard Science in the News*, Jan. 2023.

Skills

Languages R, Julia, Python, Java, SQL, $\text{\LaTeX}$, markdown, HTML/CSS
Tools Git, GitHub, Quarto, RStudio, VSCode, AWS, SLURM, Linux

Open-Source Contributor Credits

- The Book of Statistical Proofs, <https://statproofbook.github.io/>
- TMLE.jl, <https://github.com/TARGENE/TMLE.jl>
- DensityRatioEstimation.jl, <https://github.com/JuliaML/DensityRatioEstimation.jl>
- {nadir}, <https://github.com/ctesta01/nadir>
- {RieszCML}, <https://github.com/nshlab/RieszCML>